MIMO Model Predictive Controller Design for a Twin Rotor Aerodynamic System

Rajashree Raghavan, Susy Thomas
Department of Electrical Engineering, National Institute of Technology Calicut, India.

***Abstract*—Model Predictive Control (MPC) is a viable control strategy for Multi-Input Multi-Output (MIMO) systems owing to its capability of handling multivariable interactions and constraints on input, output as well as state. MPC design for MIMO systems generally resort to decoupling the system and adopting the Single-Input Single-Output (SISO) design approach. In this paper, the SISO MPC design approach is extended to a MIMO system without compromising on the system dynamics by decoupling it. The effectiveness of the designed controller is validated in simulation using a benchmark MIMO system, the Twin Rotor Multi-Input Multi-Output System (TRMS). The control objective is to stabilize the system in a coupled condition and make its beam to track a specified reference trajectory accurately or reach desired positions in two degrees-of-freedom (2 DOF). Simulation results in presence of external disturbances and parameter variations are presented to corroborate the robustness and effectiveness of the control strategy.**

***Index Terms*—Constrained Control, Cross-coupling, MIMO System, Model Predictive Control, Twin Rotor Multi-Input Multi-Output System.**

I. Introduction

Model Predictive Control (MPC), an emerging control technique has attracted much attention due to its capability to control constrained systems. The basic idea of MPC is to use an explicit model of the system to predict its future behavior and to repeatedly calculate a sequence of control signals through an optimization method in which the discrepancy between the predicted system trajectory and the desired reference trajectory over a finite time horizon is minimized [1,2]. MPC was first applied successfully for plants with slow dynamics [3]. It is now one of the most widely used multivariable control algorithms in diverse areas such as advanced manufacturing, food processing, robotics, power electronics, automotive and aerospace applications [4]. Multivariable systems have posed unprecedented challenges to control engineers due to their coupled dynamics. As MPC can handle multivariable interactions and physical system constraints within the optimization framework, it is most suited to MIMO systems. In addition, MPC is able to ensure good tracking performance, closed-loop stability and robustness.

In this context, the methodology reported in the literature for SISO systems is suitably extended to a MIMO system; the system being chosen is the Twin Rotor Multi-Input Multi-Output System (TRMS). TRMS is a laboratory test-rig developed by Feedback Instruments Ltd for control experiments [5]. It is the prototype of a helicopter and retains essential helicopter features such as system nonlinearities and cross-coupled modes. But contrary to most flying helicopters, the angle of attack of the rotors is fixed and the aerodynamic forces are controlled by varying the speeds of the motors. Research interest in laboratory set-up platforms for simulating complex aircraft manoeuvres and investigating different control paradigms has grown over the years. Many strategies to control the TRMS ranging from classical to advanced and intelligent techniques have been developed [6-13]. In [6] a robust deadbeat control technique is applied to the vertical and horizontal subsystems of the TRMS by considering the cross-coupling terms as disturbances to each other. A similar control strategy is discussed in [7] wherein a decoupled time optimal robust controller is presented. A comparison between classical control techniques and intelligent control techniques based on fuzzy logic and genetic algorithms is presented in [8] and the hardware implementation of the same in [9]. In [10], an evolutionary computation method based on real-coded genetic algorithms for the parameter optimization of PID control to the TRMS is discussed. Parallel distributed fuzzy LQR controllers are designed in [11] to control the position of pitch and yaw angles to cover various operating regions. In [12], a fuzzy sliding and fuzzy integral sliding controller is designed to control the position of pitch and yaw angles of the TRMS. A second-order sliding mode controller is proposed for position control of the TRMS in [13] wherein the system is pseudo decoupled into vertical and horizontal subsystems and coupling effects are considered as subsystems' uncertainties and perturbations. Most of these works deal with the linearized model for controlling TRMS as a decoupled system using various methodologies which naturally lead to some form of compromise in the system dynamics. In this paper, design of MPC for TRMS for achieving trajectory tracking control is reported. The system is considered in its true coupled form and also constraints on the control input are incorporated. The advantage lies in the simplicity of design without compromising on the performance requirements and also ensuring robustness.

The paper is structured as follows: In Section II, the TRMS is described and its mathematical model is presented. Section III is devoted to a brief discussion of the control scheme used. Section IV details simulation results and conclusions are drawn in Section V.

II. System Description and Modelling

A. System Description

TRMS is a benchmark example of a multivariable system representing a flexible manoeuvring structure that resembles

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a helicopter in many aspects. The system characteristics such as strong cross-coupling between horizontal and vertical axes and nonlinearities present formidable challenges in modelling and control design. TRMS has two propellers driven by DC motors at both ends of a beam which is pivoted on its base in such a way that it can effect rotation both in the horizontal and vertical planes. The main rotor produces a lifting force allowing the beam to rise vertically (pitch angle) whereas the tail rotor is used to control the beam to turn to the left or to the right (yaw angle). A counterweight fixed to the beam is used for balancing the angular momentum in a stable equilibrium position. The rotation of the tail motor slightly affects pitch angle whereas the rotation of the main motor strongly affects not only pitch angle but also yaw angle. This explains the cross-coupling effect. The control inputs of the system are the supply voltages of DC motors. A change in the voltage value results in a change of the rotation speed of the propeller which results in a change in the corresponding position of the beam. The outputs of the system are the positions of the beam, which are the pitch and yaw angles. A schematic diagram of the TRMS [14] is shown in Fig. 1. and the characteristic parameters are tabulated in Table I. The system dynamics is presented in the next section.

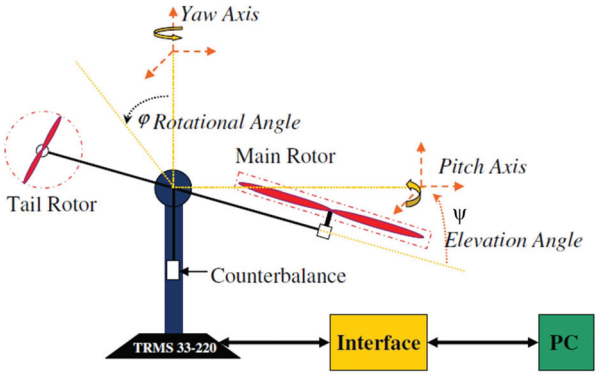


Fig. 1: Twin Rotor Multi-Input Multi-Output System

B. Modelling of TRMS

The momentum equation for the vertical movement is given as:

$$I_1\ddot{\psi} = M_1 - M_{FG} - M_{B\psi} - M_G \tag{1}$$

where the nonlinear main propeller thrust is approximated by:

$$M_1 = a_1\tau_1^2 + b_1\tau_1 \tag{2}$$

gravity momentum

$$M_{FG} = M_g sin\psi \tag{3}$$

friction forces momentum

$$M_{B\psi} = B_{1\psi}\dot{\psi} - B_{2\psi}sin(2\psi)\dot{\phi}^2 \tag{4}$$

and gyroscopic momentum

$$M_G = k_{gy}M_1\dot{\phi}cos\psi \tag{5}$$

TABLE I: Parameters of the TRMS [5]

Symbol	Description	Value
I_1	moment of inertia of vertical rotor	0.068 kgm^2
I_2	moment of inertia of horizontal rotor	0.02 kgm^2
a_1	static characteristic parameter	0.0135
b_1	static characteristic parameter	0.0924
a_2	static characteristic parameter	0.02
b_2	static characteristic parameter	0.09
M_g	gravity momentum	0.32 Nm
$B_{1\psi}$	friction momentum function parameter	0.006 Nm-s/rad
$B_{2\psi}$	friction momentum function parameter	0.001 Nm-s/rad
$B_{1\phi}$	friction momentum function parameter	0.1 Nm-s/rad
k_{gy}	gyroscopic momentum parameter	0.05 s/rad
k_1	motor 1 gain	1.1
k_2	motor 2 gain	0.8
T_{11}	motor 1 denominator parameter	1.1
T_{10}	motor 2 denominator parameter	1
T_{21}	motor 1 denominator parameter	1
T_{20}	motor 2 denominator parameter	1
T_p	cross reaction momentum parameter	2
T_0	cross reaction momentum parameter	3.5
k_c	cross reaction momentum gain	-0.2

The main DC motor with the electrical control circuit is given by:

$$\tau_1 = \frac{k_1}{T_{11}s + T_{10}}u_1 \tag{6}$$

The momentum equation for the horizontal movement is given as:

$$I_2\ddot{\phi} = M_2 - M_{B\phi} - M_R \tag{7}$$

where the nonlinear main propeller thrust is approximated by:

$$M_2 = a_2\tau_2^2 + b_2\tau_2 \tag{8}$$

friction forces momentum

$$M_{B\phi} = B_{1\phi}\dot{\phi} \tag{9}$$

and cross-reaction momentum

$$M_R = k_c\frac{T_0 s + 1}{T_p s + 1}M_1 \tag{10}$$

Further, the tail DC motor with the electrical control circuit is given by:

$$\tau_2 = \frac{k_2}{T_{21}s + T_{20}}u_2 \tag{11}$$

As in [15], a linearized model of the TRMS is considered for the controller design. The plant has 2 inputs, 2 outputs and 7 states. The inputs are the DC voltages of tail and main motors (u_h) and (u_v) respectively and outputs are the yaw (ϕ) and pitch angles (ψ) of the beam. The state variables are the horizontal position of TRMS beam (ϕ), the vertical position of the beam (ψ) , yaw rate ($\dot{\phi}$), pitch rate ($\dot{\psi}$), the main rotor momentum (τ_1) the tail rotor momentum (τ_2) and cross-reaction momentum (M_R).

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III. MODEL PREDICTIVE CONTROLLER DESIGN

MPC refers to a category of control approaches which use an explicit model of the plant to optimally calculate manipulated variables through an optimization method. For the design, the system is considered in its state-space form:

$$\begin{cases} x(k+1) & = Ax(k) + Bu(k) \\ y(k) & = Cx(k) \end{cases} \tag{12}$$

where $x(k) \in \Re^n$ is the state vector, $u(k) \in \Re^r$ and $y(k) \in \Re^p$ are the input and output vectors respectively. The A, B and C matrices have dimensions n×n, n×r and p×n respectively. The A, B, C matrices for the TRMS are as given in [15]. Since the main and tail motor voltages of the TRMS are constrained by bounds, the following bounds on the control variables are introduced: $u_{min} \leq u(k) \leq u_{max}$ where u_{min}= -2.5V and u_{max}= +2.5V as specified by the manufacturer.

The next step in the MPC design is to form an augmented model using matrices A, B and C and a new state variable vector x(k)= $[\Delta x(k)^T,\ y(k)]^T$. The augmented model for TRMS with n=7 and p=2 is [16]:

$$\begin{cases} \begin{bmatrix} \Delta x(k+1) \\ y(k+1) \end{bmatrix} & = \begin{bmatrix} A & 0^T \\ CA & I_{p\times p} \end{bmatrix} \begin{bmatrix} \Delta x(k) \\ y(k) \end{bmatrix} + \begin{bmatrix} B \\ CB \end{bmatrix} \Delta u(k) \\ y(k) & = \begin{bmatrix} 0_{p\times n} & I_{p\times p} \end{bmatrix} \begin{bmatrix} \Delta x(k) \\ y(k) \end{bmatrix} \end{cases} \tag{13}$$

where 0^T is a n×p null matrix. Δu(k)and Δx(k) are the incremental control and state variables respectively given as:

$$\Delta u(k) = u(k) - u(k-1)$$
$$\Delta x(k) = x(k) - x(k-1)$$

By using the augmented system model of (13) and a set of future incremental control parameters, the predicted state and output variables are calculated sequentially for N_p number of samples, N_p being the prediction horizon. It is assumed that all state variables are available through measurement. The future incremental control trajectory is denoted as: Δu(k),Δu(k+1),Δu(k+2).....Δu(k+N_c-1) where N_c is the control horizon.

The predicted state and output variables can be represented as:

$$\begin{cases} x(k+m|k) & = A^m x(k) + \sum_{j=0}^{m-1} A^{m-j-1} B\Delta u(k+j) \\ y(k+m|k) & = CA^m x(k) + \sum_{j=0}^{m-1} CA^{m-j-1} B\Delta u(k+j) \end{cases} \tag{14}$$

where x(k+m|k) and y(k+m|k) represent the future state and output respectively, at sampling instant (k+m), computed using current state information x(k). It is assumed that $N_c \leq N_p$ and Δu(k+N_c+j)=0 where j=0,1,2...N_p-N_c-1.

As per the methodology, the system is to track a reference trajectory with minimum error and minimum control effort. The cost function that reflects the control objective is formulated as:

$$J(k) = \sum_{m=1}^{N_p} \parallel y(k+m|k) - y_{ref}(k+m) \parallel_Q^2 + \sum_{m=0}^{N_c-1} \parallel \Delta u(k+m|k) \parallel_R^2 \tag{15}$$

where k denotes the current sampling instant, y(k+m|k) is the predicted output vector at time (k+m) depending on measurements up to time k, y_{ref}(k+m) is the reference trajectory. Q and R are positive definite weighting matrices of tracking error and control effort respectively. Minimizing the performance measure in (15) leads to a sequence of control moves. According to receding horizon principle, only the first 'r' samples of this sequence are taken to form the incremental optimal control while the rest of the sequence is ignored. At the next sampling instant (k+1), there occurs a new measurement of the process outputs and the whole procedure is repeated, with the prediction horizon of the same length Np, but shifted one step forward.

IV. RESULTS AND DISCUSSIONS

The capability of the TRMS with MPC to follow various reference signals is verified through simulations. The reference signals chosen are:

1) step input with amplitude 0.78 rad as pitch angle reference and step input with amplitude 1 rad as yaw angle reference.
2) sine wave having amplitude 0.78 rad and frequency 0.02 rad/sec as pitch angle reference and sine wave having amplitude 1 rad and frequency 0.025 rad/sec as yaw angle reference.
3) square wave having amplitude 0.78 rad and frequency 0.04 rad/sec as pitch angle reference and square wave having amplitude 1 rad and frequency 0.04 rad/sec as yaw angle reference.

The parameters of the controller are listed in Table II. The pitch and yaw angle responses of TRMS along with the control signals for step input are plotted in Fig. 2. Table III summarizes the transient performance of the TRMS using the designed controller for the step input case. The choice of the

TABLE II: MPC parameters

Parameter	Value
N_p	12
N_c	8
Sampling Time, T_s	1 sec
$u_{v,min}$	-2.5 V
$u_{v,max}$	+2.5 V
$u_{h,min}$	-2.5 V
$u_{h,max}$	+2.5 V
Q	$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$
R	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

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TABLE III: Transient performance of TRMS for step input

Time Response Specifications	Vertical (Pitch Angle)	Horizontal (Yaw Angle)
Rise Time (seconds)	6.97	2.85
Settling Time (seconds)	19.89	8.18
Peak Overshoot (%)	1.53	7.33
Control Input (Volts)	min: 0.812 max: 2.46	min: -0.73 max: 0.29

MPC parameters allow the controller to track the reference trajectories with less than 10% overshoot and a settling time of less than 20 seconds.There is no steady-state error and the control inputs are within limits as is evident from Fig. 2 and Table III. The oscillations in the pitch angle response are due to the gravity effect of the TRMS. The performance

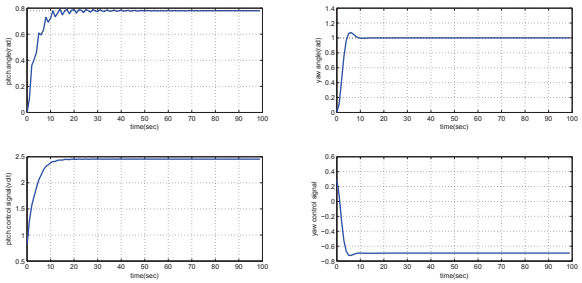


Fig. 2: Step response of TRMS using MPC

of the controller with sine and square reference signals are depicted in Figs. 3 and 4 respectively. The capability of the controlled system in following a continuously varying reference trajectory is established through these figures.

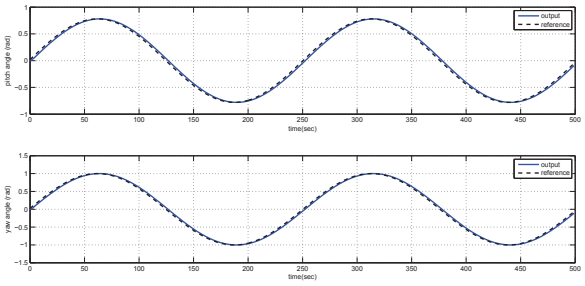


Fig. 3: Sine wave response of TRMS using MPC

As the controller is robust by design, this property is verified by introducing parameter variations viz, moment of inertia of main and tail rotors are increased and decreased respectively to 50% of their nominal values. The trajectory tracking responses in the presence of plant perturbations with respect to step, sine and square reference inputs are shown in Figs. 5 to 7 respectively. It is seen that the system remains invariant to parameter variations.

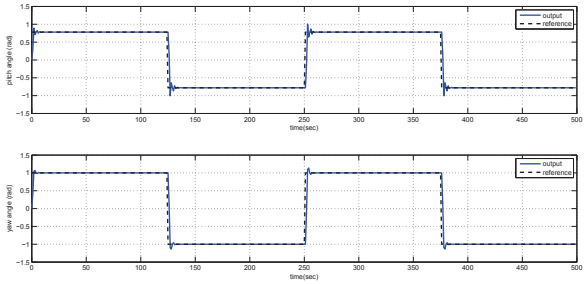


Fig. 4: Square wave response of TRMS using MPC

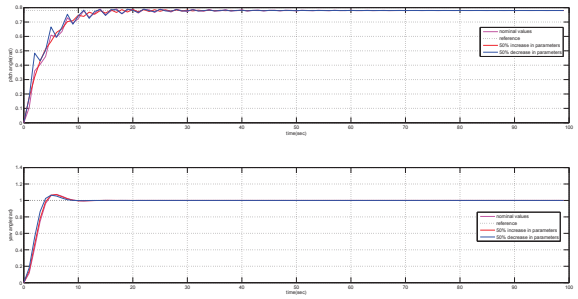


Fig. 5: Step response of the perturbed system using MPC

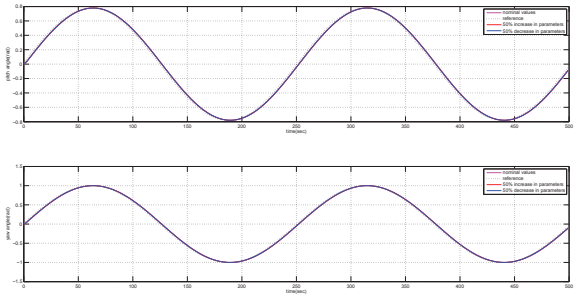


Fig. 6: Sine wave response of the perturbed system using MPC

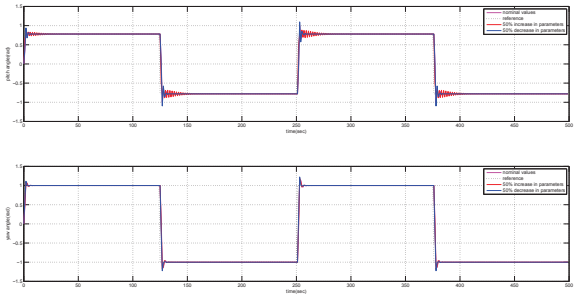


Fig. 7: Square wave response of the perturbed system using MPC

The disturbance rejection capability of the controller is tested by applying an external disturbance to the system. An

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instantaneous disturbance of 0.2 rad is introduced for both pitch and yaw angles at t=50 seconds and the corresponding result is shown in Fig. 8. The step response of the closed-loop system with an external sinusoidal disturbance of 0.1 sin(t) is shown in Fig. 9. Results demonstrate that the yaw and pitch angles deviate from their desired positions due to the disturbance applied. However, the controller rejects the disturbance and reverts the system back to its desired angles.

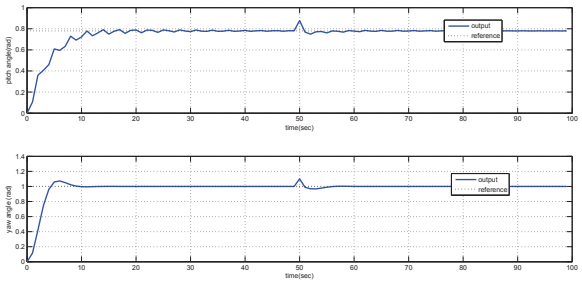


Fig. 8: Step response of the system with an output disturbance, d=0.2 rad

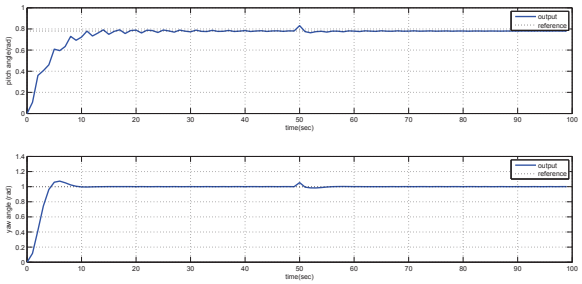


Fig. 9: Step response of the system with an output sinusoidal disturbance

V. CONCLUSION

A controller design based on model predictive control strategy has been adopted for trajectory tracking of the TRMS which is inherently a coupled system. The methodology does not involve system decoupling and hence does not approximate the system dynamics. The design is such that the controller is capable of handling the cross-coupling effects, rejecting the disturbances and maintaining robustness to parameter variations. All the traits of the controller are verified through simulations involving the tracking of various reference signals, viz.step, sine and square.

REFERENCES

[1] E. F. Camacho and C. Bordons, Model Predictive Control, *2nd edition, Springer, London* ,2004.

[2] J.M. Maciejowski, Predictive Control with Constraints, *Harlow, England: Prentice Hall* , 2002.

[3] S.J. Qin and T.A. Badgwell, "A survey of Industrial Model Predictive Control Technology," *Control Engineering Practice*, Vol. 11, pp. 733-764, 2003.

[4] Yu-Geng Xi, De-Wei Li and Shu lin, "Model Predictive Control- Status and Challenges," *Acta Automatica Sinica*, Vol. 39, no.3, pp. 222-236, March 2013.

[5] Twin Rotor MIMO System 33-949S User Manual, *Feedback Instruments Ltd., East Sussex, UK*, 2006.

[6] P. Wen and T.W. Lu, "Decoupling Control of a Twin Rotor MIMO System using Robust Deadbeat Control Technique," *IET Control Theory and Applications*, Vol.2, pp. 999-1007, 2008.

[7] Te-wei Lu and Peng Wen, "Time Optimal and Robust Control of Twin Rotor System," *Proceedings of the IEEE International Conference on Control and Automation, Guangzhou, China*, pp. 862-864, 2007.

[8] J.G. Juang, R.W. Lin and W.K. Liu, "Comparison of Classical Control and Intelligent Control for a MIMO System," *Appl. Math. Comput.* , 205, pp.778-791, 2008.

[9] J.G. Juang, W.K. Liu and R.W. Lin, "A Hybrid Intelligent Controller for a Twin Rotor MIMO System and its Hardware Implementation," *ISA Trans*, 50, pp. 609-619, 2011.

[10] J.G. Juang, M.T. Huang and W.K. Liu, "PID Control Using Presearched Genetic Algorithms for a MIMO System," *IEEE Trans. Syst. Man Cybern:* , Part C 38, pp. 716-727, 2008.

[11] C.W. Tao, S. Taurb and Y.C. Chen, "Design of a Parallel Distributed Fuzzy LQR Controller for the Twin Rotor Multi-Input Multi-Output System," in *Fuzzy Sets Systems*, pp. 2081-2103, 2010.

[12] C.W. Tao, J.S. Taur, Y.H. Chang and C.W. Chang, "A Novel Fuzzy Sliding and Fuzzy Integral Sliding Controller Design for the Twin Rotor Multi-Input Multi-Output System", in *IEEE Transactions on Fuzzy Systems*, Vol. 18, pp.1-12, 2010.

[13] S.Mondal and C.Mahanta," Adaptive Second-order Sliding Mode Controller for a Twin Rotor Multi-Input Multi-Output System," *IET Control Theory*, Vol.6, Iss.11, pp. 2157-2167, 2012.

[14] S.F. Toha, S. Julai and M.O. Tokhi, " Ant Colony Based Model Prediction of a Twin Rotor System," *Procedia Engineering*, 41, pp. 1135-1144, 2012.

[15] Jatin Kumar Pradhan and Arun Ghosh, " Design and Implementation of Decoupled Compensation for a Twin Rotor Multiple-Input and Multiple-Output System," *IET Control Theory Appl.*, Vol.7, Iss.2, pp. 282-289, 2013.

[16] Luiping Wang, Model Predictive Control System Design and Implementation using MATLAB, *Springer*, 2009.

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